

Missing Number

You are given all numbers between $1, 2, \dots, n$ except one. Your task is to find the missing number.

Input

The first input line contains an integer n .

The second line contains $n-1$ numbers. Each number is distinct and between 1 and n (inclusive).

Example

Input: 5

2 3 1 5

Output: 4

Approach

We will create a function to find the missing number using the sum of the natural numbers formula. First we will calculate the total sum of the first N natural numbers using formula $n * (n + 1) / 2$. Now we calculate sum of all elements in given array. Subtract the total sum with sum of all elements in given array and return the missing number.